

# Ahhyun Lucy Lee

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## Education

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| <b>Seoul National University</b><br>Integrated M.S. and Ph.D. Program in Brain and Cognitive Sciences<br>Seoul, Korea                                           | Sep '24 – Current |
| <b>Princeton University</b><br>Visiting Student Research Collaborator, Department of Psychology<br>Princeton, NJ, USA                                           | Apr '26 – Sep '26 |
| <b>Pohang University of Science and Technology (POSTECH)</b><br>Bachelor of Science in Life Science, Minor in Computer Science and Engineering<br>Pohang, Korea | Mar '19 – Aug '24 |
| <b>University of Illinois Urbana-Champaign (UIUC)</b><br>Non-degree Exchange Semester, Department of Psychology<br>Champaign, IL, USA                           | Jan '23 – May '23 |

## Publications

† First author    \* Corresponding author

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| <b>BAT-CLIP: Trimodal Alignment of Brain, Audio and Text</b><br>Suhyun Kim <sup>†</sup> , Jinmo Han <sup>†</sup> , Danny Dongyeop Han <sup>†</sup> , Ahhyun Lucy Lee, Jewoon Lee, Yonghyeon Gwon, Zach Paris, Chun Kee Chung, Saewoong Bahk, Nam Soo Kim, Seong Jae Hwang, Jiok Cha <sup>*</sup>                                                                                                  | IEEE MLSP 2026<br>(accepted) |
| <b>DIVER-1 : Deep Integration of Vast Electrophysiological Recordings at Scale</b><br>Danny Dongyeop Han <sup>†</sup> , Yonghyeon Gwon <sup>†</sup> , Ahhyun Lucy Lee <sup>†</sup> , Taeyang Lee, Seongjin Lee, Jubin Choi, Sebin Lee, Jihyun Bang, Seungju Lee, David Keetae Park, Shinjae Yoo <sup>*</sup> , Chun Kee Chung <sup>*</sup> , Jiok Cha <sup>*</sup><br><a href="#">Paper Link</a>  | Under review                 |
| <b>DIVER-0: A Fully Channel Equivariant EEG Foundation Model</b><br>Danny Dongyeop Han <sup>†</sup> , Ahhyun Lucy Lee <sup>†</sup> , Taeyang Lee <sup>†</sup> , Yonghyeon Gwon <sup>†</sup> , Sebin Lee <sup>†</sup> , Seongjin Lee <sup>†</sup> , David Keetae Park, Shinjae Yoo <sup>*</sup> , Jiok Cha <sup>*</sup> , Chun Kee Chung <sup>*</sup><br>preprint <a href="#">arXiv:2507.14141</a> | 2025                         |
| <b>Exploring the diagnostic potential of EEG theta power and interhemispheric correlation of temporal lobe activities in Alzheimer's Disease through random forest analysis.</b><br>Ahhyun Lucy Lee <sup>†</sup> , Eunjin Hwang <sup>*</sup> & Jeongeun Hwang <sup>*</sup><br><i>Computers in biology and medicine</i> , 192, 110248. DOI: <a href="#">10.1016/j.combiomed.2025.110248</a>        | 2025                         |
| <b>NeuroBabyLM: A Benchmark for Brain Alignment in Language Models Trained on Developmentally-Plausible Datasets</b><br>Ahhyun Lucy Lee <sup>†</sup> , Danny Dongyeop Han <sup>†</sup> , Nikunj Agarwal, Kijung Son, Jiok Cha <sup>*</sup><br><a href="#">Poster</a>                                                                                                                              | CCN 2026 (Poster, accepted)  |
| <b>Few-Shot Deep Learning Reveals Brain-Based Markers of PTSD of Sexually Assaulted Adolescents</b><br>Seoyoung Lim <sup>†</sup> , Ahhyun Lucy Lee, Ana Carolina Coelho Milani, Jiok Cha <sup>*</sup>                                                                                                                                                                                             | In progress                  |

## Enhancing psychiatric disorder diagnosis: integrating multilabel metadata with contrastive learning on T1 structural MRI

*In progress*

Danny Dongyeop Han<sup>†</sup>, Ahhyun Lucy Lee<sup>†</sup>, Junbeom Kwon, Jiook Cha<sup>\*</sup>

## Conferences

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### DIVER-0 : A Fully Channel Equivariant EEG Foundation Model

*ICML 2025 Workshop  
on GenBio (Spotlight)*

Danny Dongyeop Han<sup>†</sup>, Ahhyun Lucy Lee<sup>†</sup>, Taeyang Lee<sup>†</sup>, Yonghyeon Gwon<sup>†</sup>,  
Sebin Lee<sup>†</sup>, Seongjin Lee<sup>†</sup>, David Keetae Park, Shinjae Yoo<sup>\*</sup>, Jiook Cha<sup>\*</sup>, Chun Kee Chung<sup>\*</sup> [arXiv:2507.14141](#)



### Enhancing psychiatric disorder diagnosis: integrating multilabel metadata with contrastive learning on T1 structural MRI

*OHBM2025 (Poster)*

Danny Dongyeop Han<sup>†</sup>, Ahhyun Lucy Lee<sup>†</sup>, Junbeom Kwon, Jiook Cha

[Poster](#)

### Exploring the Diagnostic Potential of EEG Theta Power and Interhemispheric Correlation of Temporal Lobe Activities in Alzheimer's Disease through Random Forest Analysis

*SMB 2024 (Poster)*

Ahhyun Lucy Lee<sup>†</sup>, Eunjin Hwang<sup>\*</sup> & Jeongeun Hwang<sup>\*</sup>

[DOI: 10.1016/j.j.combiomed.2025.110248](#)

### The Impact of Low Motion Smoothness on Auditory Processing

*ICMPC17 (Oral)*

Ahhyun Lucy Lee<sup>†</sup>, Jinhee Kim, You Jeong Hong, Kyung Myun Lee<sup>\*</sup>, Kyogu Lee<sup>\*</sup>

[Paper](#)

## Research & Internship Experience

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### Internship in Lablup (Opensource cloud computing and AI service company)

Seoul, Korea  
Jan '24 – Aug '24

- Conducted research on the topic "Exploring the Diagnostic Potential of EEG Theta Power and Interhemispheric Correlation of Temporal Lobe Activities in Alzheimer's Disease through Random Forest Analysis"

### Internship in MARG (Music and Audio Research Group, SNU)

Suwon, Korea  
Jul '21 – Aug '23

- Conducted independent research on "Effect of Beat Coordination of Simultaneous Dance Stimuli on Sound Perception in Audiovisual Dance," developing a real-world observation into a novel research question.
- Designed and implemented experimental paradigm using PsychToolbox in MATLAB and performed statistical analysis in R.
- Presented research findings at the 17th International Conference on Music Perception and Cognition (ICMPC17).

## Honors and Awards

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### Presidential Science Scholarship, Ph.D. Track

Mar '26 – Present

- National fellowship from the Ministry of Education and the Korea Student Aid Foundation (KOSAF), awarded to graduate researchers in the natural sciences and engineering identified as having the potential to become world-leading scientists.
- One of 120 graduate scholars selected nationwide in 2026.

### National Science and Engineering Undergraduate Scholarship

Mar '19 – Aug '24

- Awarded full-ride national scholarship (KOSAF) for future scientists and engineers based on high school academic excellence.
- Maintained scholarship throughout undergraduate program with consistent strong academic performance.

## Genexine Research Award

Apr '21

- Awarded to top undergraduate researchers for outstanding achievement.

## Language

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English (Professional, TOEFL ibt 107/120), Korean (Native)

## References

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### Jiook Cha

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### Seung Tae Baek

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### Joonseok Lee

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### Uri Hasson

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